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Hanping Mao^a, Francis Kumi^{ab}, Qingli Li^a & Luhua Han^a

^a School of Agricultural Equipment Engineering, Jiangsu University, Zhenjiang, People's Republic of China

^b Department of Agricultural Engineering, School of Agriculture, University of Cape Coast, Cape Coast, Ghana

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REVIEW ARTICLE

Combining X-ray computed tomography with relevant techniques for analyzing soil–root dynamics – an overview

Hanping Mao^a, Francis Kumi^{a,b}, Qingli Li^a, & Luhua Han^a

^a*School of Agricultural Equipment Engineering, Jiangsu University, Zhenjiang, People's Republic of China, and* ^b*Department of Agricultural Engineering, School of Agriculture, University of Cape Coast, Cape Coast, Ghana*

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The study of below-ground features of roots and soil and their interactions is essential for understanding the configuration and diversities in such a dynamic environment. X-ray computed tomography is recognized as a tool for visualizing the physical interactions in the soil. In many studies, it has been used as a stand-alone tool to describe soil and root parameters. However, in recent times, attempts to couple it with other complementary tools are gaining rapid interest among researchers. The paper therefore provides an overview of the major application of combining X-ray computed tomography with other relevant methods in analyzing the structural characteristics of roots and soil media. The relevance of using this multidisciplinary approach for unraveling the mysteries surrounding root–soil dynamic interactions is stressed. The current and future trends of such studies are also pointed out.

Keywords: computed tomography, imaging, interaction, roots, soil, X-ray.

Introduction

The soil is a complex system having diverse biological and physicochemical materials. Root development and functional processes influence the soil's structure and morphology. In recent times, interests in below-ground root–soil investigations have increased rapidly. The traditional way of destructively removing, washing and examining roots to determine the extent of growth tends to be ineffective, time consuming and disturbs soil–root interactive research. This demands the use of improved and non-destructive methods. One of the most promising techniques that has received rapid attention among researchers for the attainment of such goals is the X-ray computed tomography (Duliu 1999).

Previous research has revealed its potential of investigating soil–root structure (Duliu 1999; Perret et al. 2007; Pohlmeier et al. 2008; Quinton et al. 2009), root network and architecture (Kaestner et al. 2006; Tracy et al. 2010), soil compaction (Aravena et al. 2011; Tracy et al. 2012), soil physical pore properties (Roche et al. 2010; Munkholm et al.

2012), soil hydraulic characteristics (Tippkötter et al. 2009; Aravena et al. 2014) and root–soil contact and changes (Schmidt et al. 2012; Aravena et al. 2014). Additionally, other techniques such as magnetic resonance imaging (MRI), neutron tomography (NT), positron emission tomography (PET), electrical resistivity tomography (ERT), and scanning electron microscopy (SEM) have also been used for soil–root-related investigations (Asseng et al. 2000; Perret et al. 2000; Tumlinson et al. 2008; Nagel et al. 2009; Carminati et al. 2010; Marto et al. 2014; Wehrer et al. 2014). However, none of the above-mentioned individual methods is without limitations (Handschuh et al. 2013). For example, the CT technique usually handles only very small sample sizes and has relatively short scanning periods, making it difficult to visualize large objects under continuous dynamic conditions.

To address this, Nagel et al. (2009) hinted on the possibility of using CT alongside other image analysis techniques in identifying major soil structural features. From the insightful commentary of De Smet et al. 2012, the need for such correlative

*Corresponding author. Email: maohp@ujs.edu.cn

investigations was buttressed in addition to its announcement in excellent reviews by Cnudde and Boone (2013) and Helliwell et al. (2013). There have been several other calls for integrating CT and other relevant methods in addressing specific research questions (Tollner et al. 1987b; Carminati et al. 2010; Esser et al. 2010; Hapca et al. 2011; Garbout et al. 2012; Zarebanadkouki et al. 2012). This indicates the opening up of a new research trend in the field (Iovea et al. 2009; Moreno-Atanasio et al. 2010; Elyeznasni et al. 2012; Handschuh et al. 2013).

The objective of this paper is therefore to review the potential of coupling CT with other techniques in addressing questions associated with root–soil interactions and structural changes. There is a brief overview of the principles and guidelines of CT, followed by a discussion on the combined application of the methods based on root–soil imaging. Finally, the research advantages and limitation with reference to using such tools in imaging root–soil interactions and structural changes are reviewed. Both its current and future prospective trends are covered.

Principle and guidelines of operation of CT in soil research

The contribution of Hounsfield in developing the X-ray computed tomography in 1973 has been globally acknowledged (Hounsfield 1973). Although it was originally applied in the medical field for diagnostic imaging of tender tissues and bones, after about a decade of its invention, the first attempts in using it in soil studies were carried out (Petrovic et al. 1982; Hainsworth & Aylmore 1983).

Generally, a typical CT is composed of the source of the X-ray and the detector, with the sample being placed on a holder between them. The magnification or otherwise of the sample depends on how far it is or away from the X-ray source/or detector. An X-ray image is produced when the sample is subjected to electromagnetic radiation with very high frequency and energy having density-related characteristics. Most modern laboratory CTs allow the rotational movement of the sample between the X-ray source and the detector. The results are presented as tomographic units on each plane position after image reconstruction and mathematical manipulation with different colors for easy identification. The rest of the X-rays which do not pass through the object nor absorbed by the sample finally arrive at the detector unit. It operates using the Beer–Lambert's law which relates the incident and attenuated X-rays (respectively, I_0 and I) as well as the thickness of

the object, d , and the linear attenuation coefficient, μ , in a mathematical expression as

$$I = I_0 e^{-\mu d}. \quad (1)$$

According to Phillips and Lannutti (1997), at energies less than 1 MeV, X-ray attenuation is predominantly Compton scattering and photoelectric absorption components, hence the total linear attenuation coefficient is represented by summing the attenuation coefficients, as shown in the following equation:

$$\mu_{\text{total}} = \mu_{\text{compton}} + \mu_{\text{photoelectric}}. \quad (2)$$

One of the most important steps for acquiring quality scans is the kind of preparation done before actual scanning of samples. In principle, the object under investigation must be within the field of view for better image resolution to be realized. The scanning procedure begins by the movement of the sample into the X-ray region of the system. An X-ray is an electromagnetic waveform having its photons (Bouxsein et al. 2010) and like for most industrial and medical CT systems, which are polychromatic in nature, the operator of the system must choose the energies in the form of low, medium or high resolution to meet the need at hand. The ability to differentiate soil particles from roots is best at high energies where the attenuation is approximately proportional to the density of the material. Subsequently, the different material within the soil matrix may be segmented from one another by assigning different gray scale value to individual group of voxels (Helliwell et al. 2013). Several procedures exist but in relation to soils, the commonly used ones are the globally and locally adaptive segmentations, such as the Otsu's thresholding method. Behind the Otsu method is an underlying concept of minimizing the within-class variances of object and background voxel classes. In terms of global segmentation, 'hard' threshold values are produced based on the analysis of the image histogram representing the distribution of voxels with a given gray-level intensity. Different threshold values and methods influence physical parameters differently (Tarquis et al. 2009). However, the quality of the image produced at the end of the process is dependent on the spatial resolution used and the operator's skill and judgment to minimize experimental errors. For more information on the principle of CT, the reader is referred to the excellent review articles by Taina et al. (2008) and Helliwell et al. (2013).

Applications of CT and other techniques in soil and root studies

This section will discuss the application of CT and other imaging techniques in soil–roots studies. Since the first attempts to characterize soil properties using CT, huge progress has been made with regard to the use of the technology in soil science with extensive report focused on soil structural properties (Jegou et al. 2000, 2001; Al-Raoush & Willson 2005; Schrader et al. 2007; Tippkötter et al. 2009; Bielecki et al. 2013; Ojeda-Magana et al. 2014) and roots features (Hainsworth & Aylmore 1983; Tollner et al. 1987a; Kaestner et al. 2006; Mooney et al. 2006; Schmidt et al. 2010; Tracy et al. 2015a) and the kind of interactions existing within the soil system.

During the past few years, several researchers have attempted to apply the combination of the methods (see Table 1) ranging from root–soil interactions to deformations. The future looks promising for such applications which are often carried out based on the resolution and sensor detector limitation of the CT system and its capacity to provide the needed

information to answer the specific research question. In the following, an overview of the applications of CT in the different areas of soil–root research is presented.

Soil hydraulic and structural dynamics

The availability of water in the soil is critical for nutrient uptake in crop production (Schröder et al. 2012). One of the pioneering works of using CT in the study of water uptake roots was done by Hainsworth and Aylmore (1983) using a conventional scanner. Their work showed a clear spatial variation of water content for infiltration times of 1, 5, 10, 15 and 20 minutes. The authors described the technique as a unique tool for soil–root research, especially, in accurate three-dimensional representations of soil structural configurations. Consequently, Crestana et al. (1985) also assessed the applicability of the system for soil dynamic hydraulic properties. Their work indicated a linear correlation between X-ray attenuation and water content. Tollner et al. (1987b) and Tollner and Verma (1989) were among the first to

Table 1. Summary of selected research papers on combining CT and other techniques.

Technique(s)	Objective of study	References
CT and MRI	Soil water content	Anderson and Gantzer (1989)
	Root quantification	Metzner et al. (2015)
CT and X-ray Linescan	Bulk density and water content	Tollner (1994)
	Soil aggregate distribution	Tollner et al. (1998)
CT and NT	Fluid flow in soil aggregates	Kaestner et al. (2008)
CT and SEM	Physical and chemical maps of soils	Hapca et al. (2011)
	Root response to toxicity	Arias et al. (2010)
	Pore characterization	Sok et al. (2010)
CT and PET	Physical properties of soil	Goethals et al. (2009)
CT and MIP	Quantification of soil aggregate structure	Dal Ferro et al. (2013)
CT and modeling	Cracks in soil structure	Zhang et al. (2015)
	Soil hydraulic conductivity	Elliot et al. (2010)
	Quantitative analysis of soil pore space	Ngom et al. (2011, 2012)
	Soil aggregate geometry	San José Martínez et al. (2015)
	Pore network	Ebrahimi et al. (2013),
	Soil compaction	Jivkov et al. (2013)
		Menon et al. (2015)
		Aravena et al. (2011)
CT and endoscopy	Porous rock	
CT and digital imaging tool	Microbe/Earthworm activities in soil bioporosity	Bielecki et al. (2013)
		Pagenkemper et al. (2015)
		Capowiez et al. (2011)
		Capowiez et al. (2014)
		Auclerc et al. (2013)
		Bastardie et al. (2003)
		Hall et al. (2010)
		Soliman et al. (2010)
CT and time-resolved infrared thermography	Deformation analysis of sand	
	Soil surface crust	
X-ray PCI	Root–soil interactions	Moran et al. (2000)

have noticed the possibility of coupling CT and other imaging techniques for the determination of soil water content. They evaluated two soil types at different bulk densities. For assessing the hydraulic properties, Tollner (1994) calculated the hydraulic properties and density of a cecil sandy loam and lake-land sand using X-ray linescan and CT. Each soil was sieved with a 2-mm sieve in an air dry state and rewetted under three bulk density levels. Achieving different results with similar techniques, Tollner et al. (1998) further assessed soil aggregate size distribution of fine-silty soil, collecting a total of 12 soil samples from the field for CT scanning and analysis. After utilizing the stereological approach to obtain soil aggregate distribution, the experimental values were found to be consistent with the previous work. The authors then stressed the advantage of using CT coupled with three-dimensional volumetric computations for non-destructive scanning and analysis of soil aggregates of specific geometric mean diameters. Despite these initial attempts, the study of below-ground plant systems is still incipient compared to its corresponding above-ground traits (Bardgett et al. 2014). As a result, there has been growing interest among soil scientists to perform experimental works to quantify the structure and flow characteristics in soils using non-destructive methods, such as CT and other imaging techniques. The current trend makes it one of the largest areas of applications and looks promising in terms of the utilization of such techniques for innovative investigations.

Information on the mechanism of preferential filling of pore spaces with water enables the understanding of the structure and configuration between different soil molecules (Hopmans et al. 2004; Cui et al. 2013). Mooney and Morris (2008) showed how CT and dye tracer image analysis could be used to express the influence of soil macro-structure on water potential under different soil physical properties. In their experiments, the flow of the water characteristics was evaluated by dye flow passing through the soil macropores. They concluded that the detection of the preferential flow among other factors was dependent largely on the soil water potential and pore structure, morphology and points of granular convergence. As a result, sandy soils were observed to have absorbed more dyes than other soils. However, there is a general lack of information on the uniform characterization of the diameter limits for soil pores (Allaire et al. 2009) and it is expected that the use of compatible imaging methods such as the CT and MRI could better help in accurately predicting the minimum effective hydraulic pore structure and flow dynamics.

Although CT holds promise in providing information on pore geometry, as shown in a recent

study by Naveed et al. (2014), it is limited in its application in detecting water distribution in the soil compared to MRI and NT. Tippkötter et al. (2009) observed soil and water using microfocus computerized tomography, but they admitted the need for further proof to validate the adhesion, cohesion and shape factor effects on the soil water macropore dynamics. In modeling the relationship between hydraulic and physical properties of organic soils, Quinton et al. (2009) concluded that it is quite challenging to assess all the dynamic processes under a single-stand-alone CT experimental approach. They also reiterated that the application of CT is less efficient compared with neutron radiography and MRI in visualizing real-time water preferential flow in macroporous soils. The challenge mostly lies in differentiating the artifacts and attenuations of water from those produced by the soil components. It is often difficult to attenuate between soil physical structure and the moisture content. However, recent enhancement in computational capabilities of CT and attenuation may result in the establishment of good correlation between the soil structure and the flow properties of fluid porous media.

Until recently, most studies on the soil moisture dynamics have had limitations in terms of detailed investigations (Pohlmeier et al. 2008). Doussan et al. (2006) helped inform future research about the potential of overcoming these shortcomings by correlating light transmission system with X-ray imaging methods to model soil water root uptake from the soil. By comparing such experimental approaches, the authors proposed a means of modeling soil water content and its uptake by plants. The potential of CT and filtering algorithm to overcome the difficulties in distinguishing soil water from other components has been assessed by Laia and Cruvinel (2011). The authors found a good correlation between the X-ray linear attenuation coefficients and water content. Likewise, Luo et al. (2008) coupled industrial CT with a real-time solute transport investigation method for observing and quantifying the interaction between soil moisture and its air, and solid particles. Based on these calculations, suggestions on possible real-time visualization of soil hydrological features in both 2D and 3D were tested (Brodersen 2013). The analysis of the relationship between fluid flow and its permeability in sandy soil has been reported by Kaestner et al. (2008) by coupling synchrotron CT and the thermal neutron tomography (NT) approach (Figure 1). They showed how insightful it could be in combining neutron and X-ray-based imaging, along with image processing in porous media research. As mentioned earlier, the combination of CT with other techniques with much more sensitive

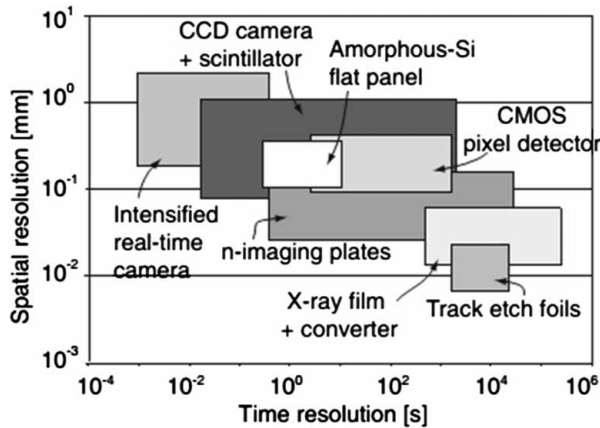


Figure 1. Neutron imaging system combined with X-ray film. (Modified from Kaestner et al. 2008).

detectors to moisture could be explored as a viable option. Recently, Bao et al. (2014) used a patterning mechanism to evaluate how plant root growth is related to the availability of water in its immediate environment.

Progress in CT as an imaging technique for discriminating, characterizing and quantifying the distribution of objects under different attenuations combined with image analysis for complex 3D analysis led to the outstanding work of Wilkins et al. (1996). In their work, the theoretical principles of X-ray radiography and PCI were utilized to develop a unique technique called X-ray absorption X-ray PCI which was used by Moran et al. (2000) to analyze pore soil structure. Porosity, pore-size distribution and pore geometry are essential material properties for investigating soil hydraulics and structure phenomena since the state of one influences the other. Gregory et al. (2010) also reported on the deformation and shrinkage effects of soil moisture with increasing pore pressure. By analyzing the decomposing changes in porosity and pore morphology, De Gryze et al. (2006) observed the addition of fresh organic composite material to soil and another sample from direct field aggregates, and, after incubation, an increase in the porosity from 3.5% to 7% in the mixed aggregate was indicated. The authors stated that this was partly as a result of the decomposition of the organic residue enhanced by soil microbial activities.

In comparison to the segmentation approach, Taud et al. (2005) developed a coherent method to determine porosity in rocky soils by calculating the CT numbers based on the attenuation coefficient of the moisture in the mean pixel volume of the slices in the 3D images. Based on the algorithm's results, they indicated an agreement with other conventional approved segmentation procedures. Gibson et al.

(2006) made a comparative assessment of the fractal analysis techniques in the assessment of scans of soil particles in terms of making 2D scans capable of truly representing the features of 3D configurations. In their study, mathematical models of soil pore–solid fractals were put forward and the CT scans analyzed to give spatial information about the soil, making a good attempt to combine theoretical and experimental techniques in soil micro-analysis. Blair et al. (2007), also used tomographic images of soil structures to validate the simulation procedure for a 3D soil up to a porosity of 11%. Changes in mechanical stress result in eventual changes in hydraulic properties of the soil (Peth et al. 2008, 2010). For example, Menon et al. (2015) used CT and modeling techniques to ascertain the impact of compaction on soil saturated water flow characteristics. They were able to reveal the drastic reduction of water flow alongside (97–99%) the total bacteria and other micro-activities in the soil by (26–39%) when its bulk density increased by approximately 60%. Soil physical properties such as mechanical strength, dry bulk density, air and water-filled pores, pH and temperature tend to affect the root architecture and its interactions within hydraulic and biotic environments (Carminati 2012; De Smet et al. 2012). Sok et al. (2010) attempted to visualize the pore characterization of soils at different scales to estimate pore structure connectivity, conductivity, permeability and using CT and two types of SEM, while Hall et al. (2010) studied the deformation of sandy soil continuum medium using the combined CT and digital imaging system. In their excellent study, Wu et al. (2011) conducted experiments using geophysical, micro-CT and reactive transport modeling methods to monitor the ureolytical-driven calcium precipitation. The nearly complete information covered by the authors gives a good indication of the advantages of combining several techniques in elucidating complex processes.

Other X-ray absorption and phase-contrast imaging-related studies have been reported by Lauridsen et al. (2014) and Van Loo et al. (2014) opening up new possibilities for further research in this area. It is possible to analyze the microstructure and link it up with the hydraulic and solid components of the soil. Future investigations in this area would be a useful tool for providing information on the structure, mechanical strength and fluid flow properties.

Besides considering the availability of soil water as the main determinant of the rate of growth of lateral roots, other factors are crucial in allowing the flow of air and other fluids within the plant system during hydropatterning (Bao et al. 2014). Torrance et al. (2008) investigated the changes in structure

and the redistribution of water to form ice lenses in saturated samples of an Aurora silt loam frost-susceptible soil. The authors proposed further study in the aspect of thoroughly providing three-dimensional continuity coupled with the detection of the displacement of hydrocarbon contaminants during freezing and thawing processes. In a recent study, Beudez et al. (2013) developed models to simulate the water flow potential of roots in relations to their capacity to take and transmit nutrients from the soil. Several models were formulated and the authors recommended combining these with the computation method for a thorough assessment of the interaction between roots and the soil.

Considering the literature on the use of the combination of CT and complementary methods to investigate the hydraulic flow in relation to the structure of the soil, quite a number of them only focus on the aspect of physical structure separately from the moisture content properties without critically making use of the full potential of such coupling studies to unravel the amazing underground root-soil-water dynamics. This calls for in-depth studies to help in filling the research gap.

Root network imaging

Soil properties within the vicinity of roots are constantly being influenced by the development of roots making the environment dynamic in nature (Carminati et al. 2010; Blossfeld 2013). The global functioning of a root system is the reflection of more or less independent physiological processes coordinated at the root system level but varying among different roots and even along a single root according to its age. This has a profound influence upon root-soil interactions and the dynamics of the rhizosphere (Hodge et al. 2009). The rhizosphere is the root's immediate surrounding where there are microbial dynamic spatial variability and interactions (Lombi & Susini 2009). Aravena et al. (2011), Daly et al. (2015) and Tracy et al. (2015b) explored the effects of root growth on the rhizosphere using CT and modeling approaches. However, their studies only centered on the physical structure as a basis to quantify the water properties of the soil. Another work has reported on provided reactive transport codes and root models (Nowack et al. 2006) and suggested that the use of these tools along with enhanced computing power could allow the monitoring of interactions between root and soil biogeochemistry and this is an area that needs further investigations in the future.

Studying roots in soil can be difficult as a result of the challenge in distinguishing it from the soil media phases by segmentation. However, we cannot

overlook the dynamic processes occurring at the root-soil interface (Mooney et al. 2012). The pioneering work of Pierret et al. (1999) in using CT to determine the *in situ* spatial distribution of chestnut and maple roots morphology in sandy and sandy clay soils was a significant contribution to using CT for root network imaging. The authors acknowledged encountering difficulties in the segmentation of the roots using low energy resolutions and as a result, to distinguish the root from the soil, different gray values were set for them. Taking a cue from Moran et al. (2000), Pierret et al. (2003) used a new X-ray-based technique made of a combination of X-ray imaging and thin slab systems to image roots and the water content and observed minor inaccuracies related to the image projection. This method also had the limitation of producing only 2D images.

In an attempt to overcome the difficulty in producing excellent 3D images of small roots, Kaestner et al. (2006) proposed the means of using a nonlinear diffusion filter and segmentation procedure under minimum noise and contrast. Using the Rosin's method, the homogenous sand media was smoothed, before the main morphological features were successfully extracted. The 3D segmented and reconstructed scans from the CT data showed a unique representation of the distribution of the roots with more than 90% of the roots in the soil system being clearly detected. This was achieved by investigating only items with a radius smaller than a given threshold before skeletonization. The experimental data-set was used to successfully validate the numerical computation performed to predict and estimate the root volume. Excellent agreement was also observed in the number of root extensions in the reconstructed image (Figure 2) and the photograph of the real roots. Nevertheless, the authors conceded that their approach was time consuming and prone to errors, and therefore encouraged combining CT and appropriate methods for segregating the root structure from the soil. Hence, in their subsequent work, Kaestner et al. (2008) combined CT and NT for in-depth study into imaging water uptake by root and their effect on the soil structure. The advantage of using CT for excellent structural images and NT for visualizing the water flow dynamics was utilized for better understanding of dynamic conditions in the soil.

Schmidt et al. (2012) described techniques which allow the estimation of contact parameters of roots extracted from 3D volumetric images of maize and lupin roots growing in cylindrical plastic containers. Image sequences were captured by segmentation and iso-surface determination tools and its accuracy was validated using physical model systems of known geometry. In a recent study, Zappala et al.

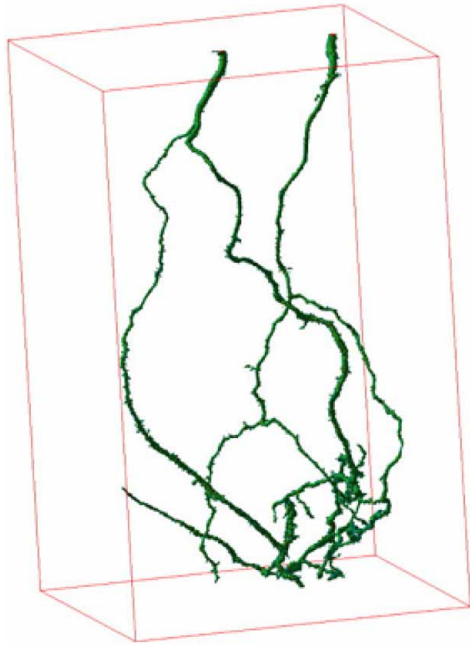


Figure 2. Visualizing images of four-month-old Alder roots from CT reconstructed data-set of dimension $36.9 \times 36.9 \times 59.15 \text{ mm}^3$ and at a resolution of $36 \times 36 \times 36 \mu\text{m}^3$ (adopted from Kaestner et al. 2006).

(2013b) demonstrated how to accurately segment roots with similar shapes at field capacity from CT-scanned data. The result was compared with that obtained from the traditional method of washing and counting the root's morphological features. The authors indicated that most of the miscalculations and errors associated with the CT segmentation of roots in soil are often as a result of soil moisture content interference. Hence, it is important to effectively reduce or totally dry the soil-root samples, before being used for scanning and segmentation. The authors (Zappala et al. 2013a) further hypothesized that X-ray doses could have a negative effect on the growth and development of the roots. However, further investigations done by the authors proved otherwise, confirming the findings of Flavel et al. (2012).

In order to obtain precise results on the effect of compaction on root growth, Tracy et al. (2012) carried out an experiment aimed at the *in situ* visualization of root-soil contact under two different compaction levels (1.1 and 1.5 g cm^{-3}) and accurate results were obtained from the analysis using ROOT-VIZ3D software. In another work, the authors presented a procedure for evaluating tomato seedling root and observed that with bulk densities of 1.2 and 1.6 g cm^{-3} , root density was greater in the more compacted region. The result also showed injuries in the form of 'swellings' at the root tip in the more compacted soils and higher tortuosity observed

being more pronounced in sandy loam soil than in clay loam (Tracy et al. 2015a).

Modeling and CT experimental work on root-soil interactions

The pioneering work of Petrovic et al. (1982) who focused on analyzing soils with respect to the spatial distribution of its bulk densities using computer-assisted tomography with excellent result of a linear relationship between X-ray attenuation coefficient and the soil's bulk density opened up CT technology to soil scientists. Tollner et al. (1987b) also explored the use of medical CT as a means for spatial observation of seedling root development and soil micro-organisms and reported a general relationship between the Hounsfield unit and the bulk density of the soil. They proposed a combination of CT with other 3D techniques for more in-depth analysis of soil systems. Other pioneering works by Anderson et al. (1988), Tollner and Verma (1989), Tollner et al. (1989), Tollner and Murphy (1991), Peyton et al. (1992) and Tollner (1994) proved the usefulness of CTs for soil-root studies and opened up research possibilities in soil science.

The characterization of the root-soil architecture is often related to shape and structure (Hodge et al. 2009). Although mostly it is challenging to separate the attenuation of root from soil medium, persistence in such efforts has led to many interesting findings for elucidating the mechanisms associated with the point-to-point interactions among different materials in the soil (Young & Crawford 2004). An important aspect of the excellent review by Lipiec et al. (2013) reiterated that root-soil interactions are also largely affected by the water deficit in the soil. There is more water uptake by lateral roots (Pierret et al. 2006), which requires energy for root system architecture nutrient transfer in the plant. The study of the water absorption by plant roots is still an open field in terms of modeling and the use of CT. One of the main targets of root-hydraulic research and investigations is the estimation of the growth, distribution and dynamic interactions. Doussan et al. (2006) improved and validated their root-water model (Doussan et al. 1998) by incorporating the actual architecture and hydraulic property parameters into it and carrying out experimental work using X-ray imaging and light transmission techniques. Couvreur et al. (2012) subsequently looked into modeling root water transfer from the soil to the plant by fully taking into consideration the root system architecture and its water conductance properties. This was reported to have heavily reduced the computational time which was a major drawback in the work of the Doussan model, making it a major groundbreaking finding in root-water uptake

modeling. In a recent work, Javaux et al. (2013) reviewed the models of Doussan and Cobreur and proposed a more detailed method of predicting water uptake by plant in the soil. Their models had parameters related to the soil and root resistance, stress, activity and hydraulic properties. However, they recommended the coupling of non-invasive imaging techniques as an excellent step in characterizing soil–root features and for obtaining useful information for validating such models. In another study, Jahnke et al. (2009) coupled MRI with PET to explore the possibility of critically monitoring the structure and mode of water and nutrient transportation dynamics of roots and other parts of the plant system.

For many years, most experiments have concentrated on developing experimental models without comparison with theoretical models. The study of compaction and water flow within the rhizosphere using a combination of CT imaging and numerical simulation reported by Aravena et al. (2011) has shown to be one of the most robust means of coupling CT with numerical simulations. In their study, the initial root structure could extract water up to 21% in the compacted soil close to the rhizosphere, due to dynamic interactions taking place with high spatial and temporal variability. In addition, in 2014, the authors (Aravena et al. 2014) adopted the same approach to quantify the effect of deformation and fluid flow in the rhizosphere by performing simulations at the normal stress range of 100–400 kPa (Aravena et al. 2014). The experimental results indicated that measured rhizosphere compaction by roots via localized soil compaction increased the simulated water flow to the roots by 27% as compared to an uncompacted fine-textured soil of low bulk density. Very recently, Tracy et al. (2015a) used CT to examine root systems of two genotypes of tomato and the effect of introducing endogenous abscisic acid and compaction on the root growth and its corresponding structural configurations. The basis of analysis was to quantify the morphological characteristics of root by non-destructive extraction of images and segmentation. The authors developed an approach to quantify physiological parameters such as root length, diameter, surface area, volume, vertical root depth and tortuosity. However, they were unable to report on the mechanism which underpins how abscisic acid could be used to remediate the negative effect of compaction on the growth and development of roots for better yield. Nevertheless, there can be more rigorous studies to compare the effect of compaction and other physical displacements on the biological and chemical spatial distribution of element in the soil by using CT, MRI, ERT and modeling tools.

Furthermore, it is essential to note that in such a modeling approach, it is important to determine the

geometry of soil particles. The geometric modeling of the soil microstructures is a recent area of study, made possible by the improved performance of computed tomography systems. In fact, the combined use of CT and modeling is becoming one of the largest areas in the application of the technique in studying soil structure changes and fluid movement. Soil structure influences water, air and heat transfer, hence the shape and size of the peds influences its capacity to allow the transportation of fluids in the system. Instability in the soil physical structure influences the flow of fluid. The works of Ngom *et al* in 2011 and 2012, in developing a robust and coherent method for the geometric modeling of soil microstructure with specific application to pore space, was a significant contribution in terms of monitoring soil dynamics. The combination of CT experiments and modeling can also be found in the study of soil aggregate geometry (San José Martínez et al. 2015), hydraulic conductivity (Elliot et al. 2010), pore network (Ebrahimi et al. 2013; Jivkov et al. 2013) and soil and rock structure (Bielecki et al. 2013; Zhang et al. 2015).

A recent review of the methods and models for assessing the activities of roots under different soil properties by Luster et al. (2009) highlighted that non-destructive techniques such as CT and MRI have been used extensively for rhizosphere research. This is due to its ability to produce excellent 3D root system images. Hitherto, Nagel et al. (2009) proposed that using the methods individually do not provide high-quality output of root structure. The authors further demonstrated how MRI and PET could be combined to unravel C isotope-labeled photosynthate in root growth dynamics. Non-invasive and non-destructive methods have been used to show the kind of stresses and strains occurring in the soil by root growth and movement. Recent studies by Kuka and coworkers used micro-CT to analyze the microstructure of undisturbed samples taken from the field. They further investigated the visualization and quantification of roots of grass and changes in the microstructure of the soil in a 30-mm-diameter core using CT (Kuka et al. 2013a, 2013b). However, the interactions in soil media are often largely dependent on the biological and physical cycling in the system (Beare et al. 1995; Passioura 2002), and this influences the growth of roots. A comparative visualization and quantification of root systems using CT and MRI carried out by Metzner et al. (2015) indicated that both techniques perform equally well, although MRI was found to be slightly ahead in terms of producing unique root distinguishing features. The comparison between the techniques was good which proved the benefit of using a coupled CT and MRI approach. The authors indicated that

unlike the CT, which has the ability to clearly image soils of all types, MRI is limited in its operation under strong magnetic fields which could disturb the quality of the structure generated (Figure 3). Consequently, because of the limited understanding on how roots and soils behave under a range of soil physical, biological and chemical properties, especially at the micro-level, the future research direction aimed at combining such complementary methods looks promising.

Organic matter content of the soil

For the study of organic matter (OM) composition representing residues and other material decomposed in the soil, there are general challenges in distinguishing them from the main pore particles, hence for a long time, many researchers have avoided this area of investigation. One of the key issues relies on the fact that although the properties of the OM are different from the bulk soil (Kasteel et al. 2013), it is still difficult to differentiate between the features due to the similarity in attenuation between the different components of such soils (Quinton et al. 2009). However, in recent years, some attempts have been reported in overcoming this limitation.

Detection of organic matter

Due to the difficulty in CT scanning organic soils compared to natural soils, relatively few research

attempts have been made to characterize the properties of such materials. Blais (2005) assessed the physical and hydraulic characteristics of organic soil and found that the quality of CT images is influenced by the water holding capacity of the soil caused by the changes in the density of the soil matrix. An impressive study by Sleutel et al. (2008) and Quinton et al. (2009) further opened up this area of quantifying the internal architecture of the soil. Sleutel et al. (2008) combined several detectors with different sensitivities and filters at lower X-ray energy, while reducing the noise. The segmentation of the OM content from the pore particles was effectively carried out by this approach. Subsequently, peat soils, mostly organic in nature, were assessed for their OM composition by Quinton et al. (2009). They were unable to discriminate it from the moisture content, indicating that it was necessary to thoroughly dry the samples before scanning. A common way of enhancing the contrast between images of similar attenuation is by using staining agents capable of forcefully exciting the electrons on the surface of the object. In an attempt to distinguish between the linear attenuation of OM and water compositions, Kettridge and Binley (2008) used lead(II) nitrate solution to successfully increase the linear attenuation of *sphagnum* peat by flushing it over the sample before CT scanning. Biogenic gas bubbles and peat structure were identified from the CT scans of the samples. Despite the success in their work, much care should be taken in repeating

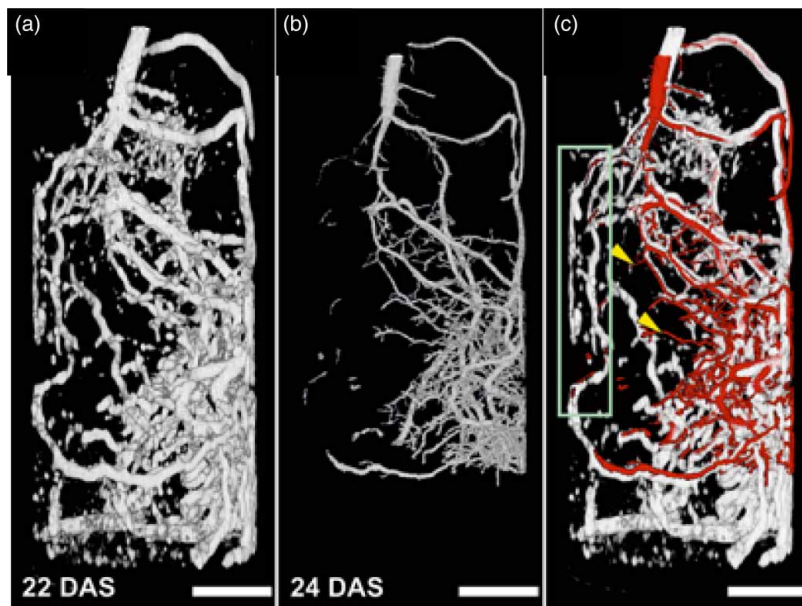


Figure 3. Three-dimensional visualization of beans roots using CT and MRI. (a) MRI image performed 22 days after sowing (DAS) at a voxel size of $330 \times 330 \times 1000 \mu\text{m}^3$. (b) CT image obtained two days after the MRI scan at a voxel size of $56 \times 56 \times 56 \mu\text{m}^3$. (c) The MRI-CT co-registered image with MRI image in gray and CT in red showing the roots. The yellow arrow and green bounding box point to the roots that were observed with CT but hidden when MRI was used. (Taken from Metzner et al. 2015).

such studies as the procedure comes with health concerns due to the carcinogenic nature of lead(II) nitrate. The pioneering study of Iltis et al. (2011) in using micro-CT for visualizing the structure and configuration of porous media using potassium iodide (KI) as a contrast-enhancing agent presented a unique method of differentiating contrast attenuation of soil aggregates and validating the result with the light microscopy approach was carried out. As a rule, its effective usage in visualizing porous media depends on the properties of the elements in the medium, as it is more useful for low-density elements (Hondo et al. 2014). Likewise, Peth et al. (2014) used osmium as a novel staining agent to distinguish low-density elements in soil OM in CT scanning. Their findings proved useful due to the ability of osmium to react with C and consequently distinguish the low and high material from other elements in the soil. However, the researchers suggested establishing the relationship between the intensity and concentration of OM content in their subsequent work.

The protocol developed by Hapca et al. (2011) on the use of 3D CT data of soils followed by 2D SEM chemical maps allows for the physical and chemical (C, O, N) component interpretation of soil opening a new line of research in the aspect of combining different techniques. This approach demonstrated how inorganic and OM compositions of soils could effectively be analyzed to obtain detailed information on samples that can be scanned repeatedly without influencing the physical processes. The higher organic content and nutrient supply enhance biological activities for improved pore connectivity by way of ensuring the quality of soil structure.

The combined use of CT and other analysis procedures is certainly based on the capacity to be complementary in obtaining detailed 3D information of spatial distribution of the organic materials. Bouckaert et al. (2013) found the compatibility of CT with other biological analysis procedures in detecting soil organic decomposition without any effect of the X-ray energy on the medium. In a recent study, Kravchenko et al. (2014) developed a new method of measuring soil particulate OM by combining image preprocessing steps with discriminant analysis classification using micro-CT. Through this procedure, they were able to analyze soil particulate OM with different gray value scales and aggregate volume distributions. However, although this and other combined techniques seem reliable and robust, they are yet to be fully utilized to gain deeper understanding into the mechanism of OM decomposition and dynamics in undisturbed soil samples. Comparing data from the experiment using leacher with that provided by non-leacher staining agents on *in situ* soils, Kasteel et al. (2013) found that the latter was more

effective in establishing the relationship between the structure of the soil and roots distribution as key parameters for quantifying the OM composition of soil.

Earthworm burrow and microbial activities

The biopore is a key area of interest in the root–soil dynamic interaction where living organisms interplay with the soil bulk media and also for the uptake of nutrient roots by the plant. In recent times, several investigations into assessing the effect of earthworm and microbes activities in these zones using the CT imaging technique have received much attention (Bruneau et al. 2005; Wuest et al. 2009; Capowiez et al. 2011; Huang et al. 2013; Helliwell et al. 2014). One advantage of such biological activities is the enhancement of root growth and pore connectivity (Dal Ferro et al. 2013). The non-destructive nature of CTs allows for real-time *in situ* monitoring of such processes. The technique has been well documented as having no significant effect on the nature and health of living organisms in the soil (Schmidt et al. 2015).

Earthworms, which are often called ‘soil engineers’, are major regulators and catalyst in OM amendment and remediation in the soil. Researchers generally classify earthworms into three groups: anecic, epigeic and endogeic and readers are referred to Taina et al. (2008) for a more detailed description of their roles and behaviors. The key roles played by earthworms impact the functioning and structure of the soil, and this is often linked to the type of earthworm and the intensity of their activities (Bastardie et al. 2003). In an exciting study to remediate the negative impact of wall compaction due to earthworm burrows, Auclerc et al. (2013) assessed the effectiveness of using limes in an acidic forest soil using CT. Although the authors failed in their attempt to use CT to investigate and to produce the expected results in the long term, their earlier application showed that ability of the lime to increase the pH up to 5.3 could contribute positively to the earthworm activities in relation to soil structure and configuration. In a similar approach, Langmaack et al. (2002) previously visualized and quantified the positive effect of increased population to combat tillage-induced compaction. Works by Schrader et al. (2007) and Pagenkemper et al. (2015) provided insightful procedures as to how CTs are combined with other complementary methods to assess the impact of earthworms on the surrounding soil. Schrader et al. (2007) used CT and SEM to describe the structural changes as a result of earthworm activities. The study revealed the evidence of 11% increase in bulk density in soils which have been burrowed by earthworms. Despite this observation, the authors

indicated that such activities have no negative effect on the microstructure of the drilosphere. In another study, Aravena et al. (2011) showed that minimum soil compaction tends to influence particle arrangements which serve aide in improving the hydraulic properties of the system. In quantifying the effect of earthworm on the structural arrangement of soil particles, Pagenkemper et al. (2015) employed CT scans to analyze the microstructure properties of the biopore. Consequently, endoscopy image acquisition for detecting the burrow wall's physical, chemical and biological characteristics as well as the sequential movement of the organisms within the system was done. The authors concluded that the earthworm movement influences the structure and properties of the surrounding soil. This is a typical example of an excellent combination of CT and other techniques to address specific questions than relying on only a single method for research. Additionally, many other efforts have been made to conduct investigations into the mechanisms and processes of earthworm–soil interaction in the drilosphere. Typically, Capowiez et al. (2014) investigated the mechanism of endogeic earthworm burrowing processes and the study revealed a general trend as the population per macropore increases the tortuosity decreases and this has a significant adverse effect on the movement of water. The work pointed out that CTs could effectively be used to assess these dynamic processes at the micro-level. Thus, such minute below-ground activities have direct influences on the overall growth and performance of the plant by providing avenues for root growth and nutrient acquisition (Newington et al. 2004; Kautz et al. 2014). However, one of the main sources of error in identifying earthworm burrow in the soil using CT is the difficulty in distinguishing between actual macropores created by the animal and those which are as a result of natural causes such as aging (Taina et al. 2008). Consequently, the visualization of other minute microbial activities tends to be relatively challenging under real-time conditions. Despite several attempts aimed at applying techniques and algorithms to assess the macro- and micro-processes in the soil, understanding the mechanism of earthworm bio-activities still remains fully unanswered. Further studies on coupling CT and other non-invasive methods are necessary to provide detailed information to help overcome this limitation in soil system dynamics.

Unlike earthworms, microbes usually absorb low attenuation, making them relatively invisible (Kuz'yakov & Blagodatskaya 2015). Recent research attention is gradually shifting towards monitoring the soil processes induced by microbial activities, as they create suitable environment for plant growth (Mooney et al. 2012). However, these activities are often

confounded by the availability or otherwise of energy and carbon elements (C). Root growth and burrowing animals are major drivers in inducing carbon into the soil horizon (Kastovska & Santruckova 2007; Ge et al. 2015). Solomon et al. (2012) combined scanning transmission X-ray microscopy (STXM) and near-edge X-ray absorption fine structure (NEXAFS) spectroscopy to assess the K- and L-edges capacity to sequester carbon and other elements in the soil. Likewise, Chen and Sparks (2015) used a similar multi-elemental approach to investigate the kind of relationship that exists between OM and mineral contents of the soil. These attempts among others open up the area of carbon (C) for developing more comprehensive models linking OM content soil mineral properties at the micro- and nano-levels. Liu et al. (2015), however, focused on the assessment of the rate at which phosphorus (P) content of the soil changes under real-time conditions using K-edge X-ray absorption near-edge structure and NMR spectroscopy under sequential fractionation. The authors showed how organic soils improved with P to boost agricultural productivity.

The 2D and 3D quantification of the edaphic structures of soils has been made possible by CT. However, until recently, CT had failed in its application to track microbial interactions and processes. Monitoring such activities involves the causal effect of microbes and the impact it has on the organizational structure and properties of the soil. In their model, Crawford et al. (2012) combined CT and the inoculation of microbes into soils to assess the interactions between microbes and soil structure. The authors reported a self-organizing concept from the soil microbial activities and reiterated the consequence of feedbacks between these processes and the microstructure of the soil irrespective of the type of microbe being investigated. The model further proved that a change in the microbial population and feedback dynamically influences the hydrological, physical and chemical properties of the soil. However, it should be noted that the experiments were conducted at constant water content allowing for clear differentiation of the attenuation of the microbes from the soil. There is therefore a need to characterize dynamic soil microbial processes using a single-spot CT method in combination with the relevant techniques to elucidate the physical and chemical processes that take place in the biosphere (including the drilosphere), rhizosphere and detritosphere.

Advantages and limitation of coupling CT with other techniques

The use of CT in a non-medical field such as soil science has gained much attention among researchers

and the future looks exciting as a research tool (van Kaick & Delorme 2005). Its ability to produce 3D images under non-destructive conditions coupled with the incorporation of filters to minimize beam hardening effects are some of the numerous advantages of CTs. Also, the convenience of estimating physical properties such as density, porosity, pore connectivity and diameter as well as its capacity to produce excellent morphologies of objects ranging from the structure of soils to monitoring the activities of soil microbes gives it an added edge revealing structural properties and bio-physical processes within the soil profile (Cnudde & Boone 2013). However, as a result of some of the earlier mentioned limitations such as image contrast (especially in water and physical soil imaging) and the fact that it operates under constant resolution per scanning session makes the means of monitoring dynamic conditions challenging using this stand-alone tool only.

Many studies have, therefore, used the combination of CT with other methods to address soil and root research questions (Table 1). It appears that this is because it allows for fast non-destructive dynamic investigation of the internal components of an object without sacrificing it. The information obtained from CT systems is often detailed enough to estimate the soil particle mineralization and system morphology and architecture. Despite improvement in the system's attenuation and throughput, this useful technique has obvious limitations such as image blurring and segmentation errors (Pierret et al. 2005). It is therefore, essential for researchers to direct efforts towards combining CT with other complementary methods to enhance cutting-edge research while industrial work on improvement of the technology continues.

The cost of CT and other systems is the main bottleneck, especially for soil research. Such facilities are only available at few institutions in single quantities, making it difficult to carry out combined research. Additionally, quantitative 3D analysis is often not straightforward and can be prone to several systematic errors caused by the effects described below. It therefore, requires careful optimizations of image quality and rigorous image analysis protocols that minimize elements of subjectivity in experimental work (Schmidt et al. 2010).

In addition to this, the use of different computer-suitable softwares and codes to analyze data of soil structure (Moreno-Atanasio et al. 2010) sometimes makes it difficult to have a general basis for assessing data analysis presented and to follow research trends. There are several software packages such as IMAGEJ, VGSTUDIO MAX, AVIZO, PORE3D and MORPHO+ currently being used, having different configurations and effectiveness in quantitative, qualitative and

statistical data analysis. Therefore, collaboration with industry is needed to evaluate and further adopt the existing effective softwares into developed systems.

Additional trends and conclusion

It is now evident that CT is a useful tool in soil research considering its enormous usage in quantifying various below-ground parameters non-destructively (Tracy et al. 2010; Helliwell et al. 2013; Zappala et al. 2013b). Although currently, there are several ongoing studies in the area of soil-root research that employ CT for visualization and quantification of essential parameters, there is still a lot of work that needs to be done with this technology. Continuous improvement in the techniques and methodologies are needed for effective soil-root research to serve as a leading imaging technology in the coming decades. Research interest in the development of theoretical models and simulations and validating such works with experimental results is an expanding area of interest. The excellent work by Aravena and coworkers has paved the way for further rhizosphere studies. Aravena et al. (2014) demonstrated how plant roots through localized mechanical stress increases the fusion of already-existing extremely low-density soil aggregates. This increases the amount of potent flow paths for water and nutrients under unsaturated conditions for constant inter-aggregate gas exchange gaps. Future work in the area of modeling and CT experimental approach could seek to identify the essential mechanism and also find out the root's ability to grow between a given inter-grain force as influenced by the history of its growth path (Wendell et al. 2012). To fully understand the internal soil environment conditions and processes, there is a need to further perform chemical (Gregory & Hinsinger 1999) and physical and micromechanical analysis as well. This could give a much broader field of investigation in terms of interactions at the root-soil interface in the future.

The development of what Pires et al. (2011) termed as third- and fourth-generation CT could be a fascinating alternative for dynamic soil-water studies, if explored in the future. It could result in the construction of models for flow processes involving hydraulic conductivity, solute infiltration, root development and help in numeric simulation of the significant soil physical dynamic processes occurring in soils. Current progress in soil research opens up efforts in applying correlations among soil properties and responses due to the activities of root and other biological living organisms aimed at addressing some of the fundamental research questions. For

example, CT and other complementary techniques will soon make it possible for full soil–root chemical and mechanical mechanisms and interactions through the cooperation and interactive research efforts from experts in fields from different backgrounds, as shown by Bao et al. (2014). The authors pointed out the potential of demonstrating mechanisms into sensing the water dynamics of root–soil processes in the future.

The analysis of images in terms of segmentation still seems to be a major hurdle since to date there is no agreement among researchers as to the standard approach for such measurements for different samples. This is possibly as a result of differences in procedures and softwares used. Suitable computer simulation softwares and codes could be freely provided and shared among researchers, through open source libraries, educational forums and conferences to enhance cutting-edge research of soil–root feature characterization.

The process of meeting future global climate and food challenges involves securing excellent plant–soil ecosystem, and this centrally depends on addressing issues related to soil–root–fluid interactions for increased yield. The aspect of modeling as a research tool in combination with CT scanning for critical analysis of role of living organisms in the soil presents an excellent opportunity for studying the effect of organisms' activities on the physical features, as well as other mechanical and biological properties within the system. In that case, strong database for linking the role of the above-mentioned parameters with the soil management and stability could be obtained. It could also be exciting in the future to incorporate the discrete-finite element modeling into CT studies to assess the dynamic states of the soil. This area is fairly unexplored and could pave the way for providing the information to understand soil–root dynamics. With much improvements in system configuration and availability of CT and other imaging tools, the provision of further insights into soil and root environments could be revealed, thereby paving the way for the optimization of soil–crop nutrition as a measure towards ensuring future food security in the coming decades.

An understanding into soil structural stresses' responses to biological, chemical and mechanical internal activities is the key to developing models for different types of soils. For example, it is important to investigate ways of changing the spatial architecture of the microbial activities and their influence in the soil media (Feeney et al. 2006). Further research attention must be geared towards developing methods for determining the relationships between air movement and entrapment under different soil conditions and the root spatial distribution

(Luo et al. 2008). Another step to addressing this is to develop systems that combine the principles of non-invasive laboratory systems with geophysical field experimental instruments to aide effective direct field non-destructive studies. Such attempts could eventually lead to elucidating the factors involved in soil structural models under dynamic conditions.

Until recently, imaging soil biota processes was described as one of the most challenging issues in the use of CT technique. Meanwhile, such critical processes which ensure the overall nutrition and structural configuration of the soil system is important to capture accurately. With current improvements in resolutions, more complex investigations into several hidden processes within the soil horizon are being carried out. There is, however, little information on the mechanism involved in the signals and activities by soil microbes. In essence, while the use of CT could be used to explore the structural effect of earthworms and other microbes on the soil, it is somehow limited in providing the detailed information on the spatial distribution of materials in the system. This calls for the combination of CT and other suitable techniques, to collect consistent data based on long-term behavior of the living organisms in the soil habitat. Hence, much needs to be done in coupling CT with other soil system monitoring techniques as the extent to which soil micro- and macro-organisms are monitored and their processes quantified is far from satisfactory.

In the future, research effort could be geared toward looking into how the biogeochemical activities in the soil influence soil properties and osmotic pressure of root hairs during water and nutrient uptake. While many studies have been done under real-time investigations, majority only limit their scope to properties of macropore and structural morphology. Essentially, in the future, we need to carefully discriminate soil physical features from the roots, water and other aspects of the system through the use of CT and other relevant techniques. Although this goes with cost and dedication of time and effort on the part of researchers, effective collaboration between research institutions and industry could help in making huge progress in this direction.

In conclusion, this review has brought to bare the possibility of combining CT and other complementary techniques and an attempt to set up a benchmark to investigate the behavior of roots and soil dynamic interactions. There is relatively little information on the mechanical and chemical aspect of root–soil interactions using CT. It is therefore essential for future research to focus more on exploring how the technology could be used to provide relevant data for the benefit of the scientific community. Based

on the expanding interest in this area, the successful use of CT and other complementary methods herein reviewed, the adoption of such combined studies should be encouraged.

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